Hospital Management System

MAD1 Project Report

1. Author

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About Me:
My name is Siddharth Sahu, a Diploma-level student in the IIT Madras BS Degree program. I am from Prayagraj and enjoy building practical, real-world web applications. I like experimenting with backend logic and creating simple, clean user interfaces.

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2. AI / LLM Usage Declaration

I used ChatGPT for approximately **5–10%** of the project.
AI was used only for idea clarification, debugging help, and improving understanding of certain implementation steps.
All database design, routing, application logic, and integration were implemented manually by me.

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3. Description

This project implements a complete **Hospital Management System (HMS)** where Admins, Doctors, and Patients can manage essential hospital operations. The system supports authentication, dashboard-based navigation, appointment workflows, treatment history, conflict-free booking, and profile management across roles.

AI/LLM usage percentage: **5–10%**, limited to conceptual help and debugging assistance.

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4. Technologies Used

Core Technologies

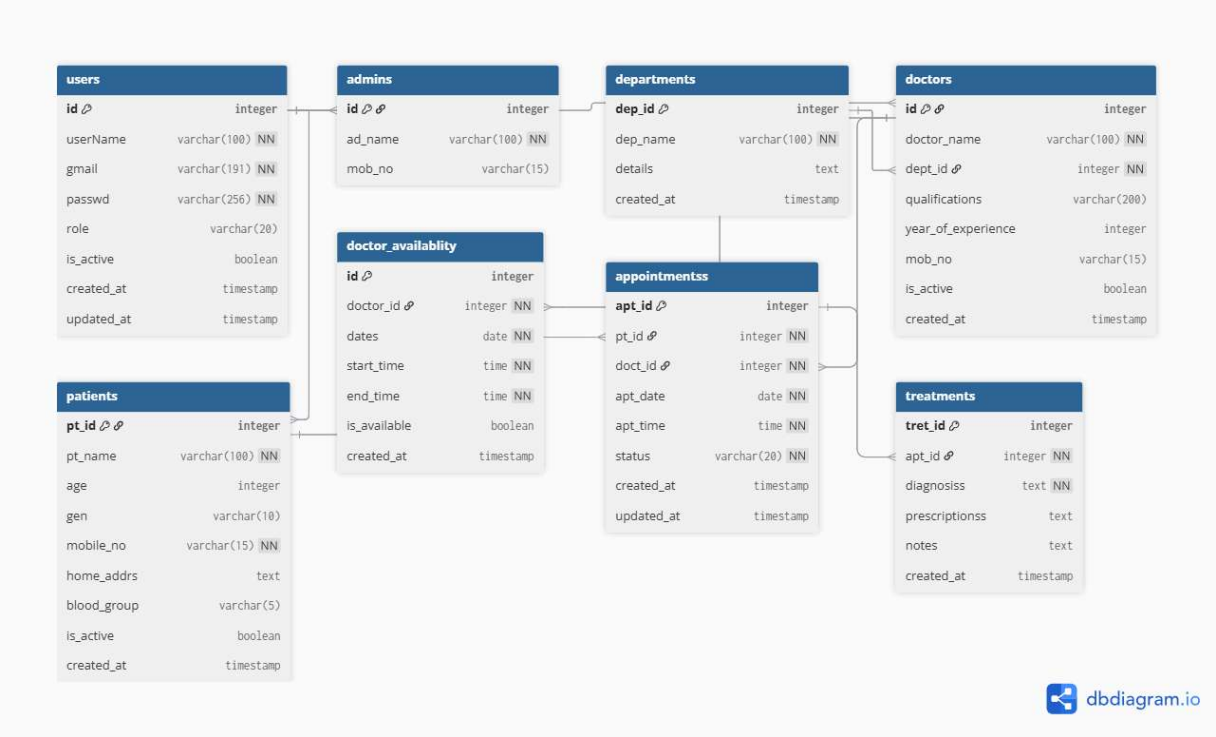
- Flask — backend routing and application logic
- Jinja2 — dynamic HTML rendering
- SQLite — lightweight database
- SQLAlchemy ORM — model definitions and relationships
- Flask-Login — authentication and role-based access control
- WTForms / HTML forms — input handling and validation
- Bootstrap 5 — responsive and clean UI
- Standard Python libraries — datetime, utilities, matplotlib(for charts) etc.

Why These Technologies?

Flask provides flexibility and simplicity suitable for the project scale.
SQLite integrates easily without external setup.
Bootstrap ensures responsive design.
Flask-Login enables secure access control.
ORM-based models guarantee proper programmatic DB creation as required by MAD-1 guidelines.

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5. Database Schema Design



Relationships

- One-to-Many → Doctor → Appointment
- One-to-Many → Patient → Appointment
- One-to-One → Appointment → Treatment
- One-to-Many → Department → Doctor

Design Justification

The schema ensures clean separation between users and medical records.
Appointments link doctors and patients, while treatment details remain tied to each appointment.
Conflict prevention (doctor + date + time) is straightforward with this structure.

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6. API Design

No separate API endpoints were implemented.
All functionality was built using standard Flask routes that render Jinja2 templates.
This meets the core requirements of MAD-1 and maintains a simple, clean architecture.

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7. Architecture and Features

Project Structure Overview

- **app.py —** main application entry point
- **models.py —** all SQLAlchemy models and DB setup
- **templates/ —** HTML templates for Admin, Doctor, Patient workflows
- **static/ —** CSS, JS, and assets
- **SQLite database** is generated programmatically on first run

The project follows an MVC-like separation:

- Models represent DB structure
- Controllers are Flask routes
- Views are Jinja2 templates

Implemented Features

Admin Features

- Login with pre-created Admin credentials
- Dashboard showing counts of doctors, patients, and appointments
- Add/update/delete doctor profiles

- Search doctors and patients
- View all appointments
- Blacklist/remove users

Doctor Features

- Dashboard with today's and upcoming appointments
- Update availability for the next 7 days
- Mark appointments as completed/cancelled
- Add diagnosis, treatment notes, and prescriptions
- Access complete patient medical history

Patient Features

- Register and log in
- Update profile
- Search doctors by specialization
- View doctor availability
- Book, reschedule, and cancel appointments
- View treatment history and past medical records

System-Level Features

- Role-based access control using Flask-Login
- Programmatic DB creation (no manual DB browser)
- Prevents double booking for the same doctor/time
- Clean and responsive UI using Bootstrap

8. Video Presentation

Video Link:
(https://drive.google.com/drive/folders/1NgRNhEVLizBWF0D6reNQIB5kJiOJP74E?usp=sharing
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